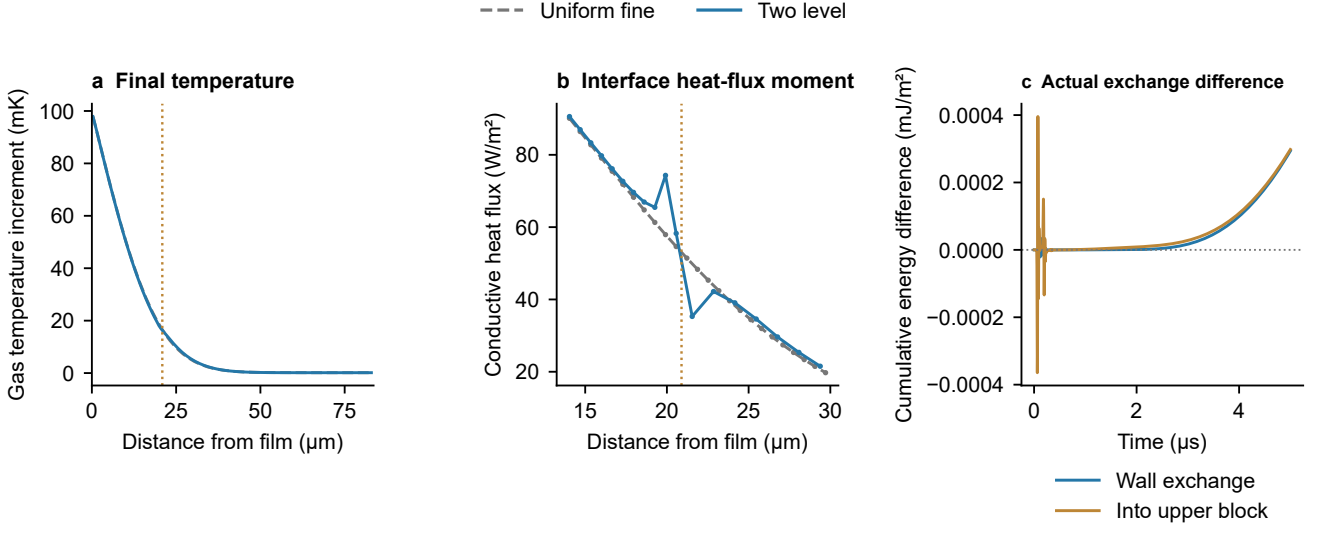
Uniform fine
Two level
a Final temperature
Gas temperature increment (mK)
100
80
60
40
20
0
0
25
50
75
Distance from film (μm)
b Interface heat-flux moment
Conductive heat flux (W/m²)
80
60
40
20
15
20
25
30
Distance from film (μm)
c Actual exchange difference
Cumulative energy difference (mJ/m²)
0.0004
0.0002
0.0000
−0.0002
−0.0004
0
2
4
Time (μs)
Wall exchange
Into upper block